



Modeling economic environment monitoring to enhance economic tasks using deep learning with economical education on various sources

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Abstract

This research study proposed a new model that consists of modeling, designing, and implementing a CNN-LSTM-based deep learning model for monitoring the economic environment to enhance economic tasks using economic education from various sources through safe information. LSTM is used because of its ability to learn long-term dependencies, in taking the concatenated function of commercial, marketing or financial data as input, while integrating encapsulations, using objects connected, smart data, and multiple social media sources to retrieve relevant safe information by combining multiple search results. It provides some good ideas extended to improve corporate marketing, business strategies, and financial times series forecasting. It is shown that the proposed model learns to both localize and recognize different aspects of marketing, business, and finance. The model evaluates the challenging task of detecting fraud in business and finance and shows that it is more accurate than the state-of-the-art of other neural networks and uses fewer parameters and less computation.

Keywords: CNN-LSTM-based multimodal deep learning model, Economic entities, Multiple social media sources, Education sustainability, Modeling, Objects connected, Smart Data

2020 Mathematics Subject Classification: Primary 03C45, 03C99, 11B34, 11B50, 11D85, 11P55, 12E12, 12L12, 13-04, 13-11, 13C20, ————— Classification, other, Representation Fct, Sequences, Repres Pbm, Equations in general fields, Model Theory of fiels, Software, sources code, research data, Claa groups,

1 Introduction

Various information on business, marketing, and finance circulates in daily life on various sources namely connected objects, smart data, and multiple social media sources with generally qualitative and significative differences in obtained information through advances in IoT to get a comprehensive view of the ongoing event. Managers are faced with a huge retrieval of information shared on the web with huge awareness: it is a great challenge.

Several artificial intelligence systems are designed for monitoring the networking environment to facilitate managers identifying and filtering relevant useful shared content. Some work is focused on using just social networks as an information source while others are concurrently dedicated to warning, multiple sources, and education.



Modeling and designing economic environment monitoring to enhance economic tasks using deep learning with economical education is a great challenge.

Research on collecting content using deep learning is a problem of sequential learning. Therefore, a new model is proposed for improving economic tasks based on a CNN-LSTM used to learn long-term dependencies. This new approach makes it possible to integrate artificial intelligence technologies, deep learning, social media and IoT technologies for managing economic tasks using safe information. Built on prior work that forms the background for this approach, this new model combines representation training with warning and economical education while integrating encapsulations from connected objects, smart data, and multiple social media sources to capture relevant and safe content while combining various search results, providing the best ideas for its extending to enhance economic tasks.

This research study consists of monitoring the economic environment to identify relevant content of an eventual upcoming economic crisis event. Once content is retrieved and cleaned of non-informative information, it is used to update content (warning or economical education) for citizens and managers to make quick and effective decisions to help people in need and avoid economic crises. Therefore, the study provides not only a solution to the challenge but also achieves promising results.

2 Results

This research study consists of monitoring the economic environment to identify relevant safe content to enhance economic tasks. Once content is retrieved and cleaned of non-informative information, it is used to update content (warning or economical education) for managers to make quick and effective decisions to enhance economic tasks.

Much research exists for Financial Time Series Forecasts. Research for Fraud Detection is common. On the other hand, there are only two studies for the Market Price and only one for Placing products in a supermarket.

Twitter is the first social network to be used in applications to exchange news via tweets. The NN-based model follows it to enhance warning that uses Twitter & Facebook.

The FFNN-based ALE follows this to enhance warning and education from the Web: it is the first to use multiple social media sources with all the Web. There are some other works namely LSTM, RNN, and Deep CNN-LSTM-based model to improve warning, awareness, and education in crisis event from all the Web.

There is the first model to use smart data & all the Web: it is the RNN-based ALE improving awareness.

With advances in IoT, other applications are used based on sensor technology. Thus, the study was able to integrate the deep learning concept.

The new proposed model consists of modeling, designing, and implementing deep learning model to enhance economic tasks through economical education while using safe information. There are other models to enhance preventing fake news, namely the improvements in commerce and business, financial time series forecasting, fraud detection and marketing with deep learning while avoiding false information, followed by Deep learning-based managing disaster with smart education in environmental sustainability while avoiding false information, followed by Smart education with deep learning and social media for managing disaster in the pursuit of environmental sustainability while avoiding false information, followed by Context-based detecting false information, and Detecting



fake information using deep CNN-BLSTM-hybrid model through context social: all these models use multiple social media sources (all the Web).

The new CNN-LSTM-based deep learning model model is presented here. The LSTM layer is powerful in handling temporal correlation. Its extension has convolutional structures in both input-state and state-to-state transitions, which will solve this problem. By stacking multiple CNN-LSTM layers and building a coding prediction structure, the research study created a network model for these space-time sequence prediction problems. The economic crisis forecasting goal uses previously observed social networking to prevent an economic event in a local region, such as Algiers, London, or Paris. From the automated learning perspective, this is a problem of predicting space-time sequences. Suppose the research study has a dynamic system represented by an $(M \times N)$ grid with M rows and N columns. In each cell of the grid, there are P measures (word, bias) varying in time. At any time, the observation can be represented by a tensor X belonging to $R^{P \times M \times N}$, with R denoting the domain of observed traits. With recording periodically observations, the study will have a sequence of tensors $X_1, X_2, \dots, X_t$. Spatio-time sequence prediction predicts the most probable sequence of length K , given previous J observations (including the current sequence) formulated.

Observing at each timestamp is a 2D map. In dividing this map into non-tiled, non-overlapping patches and visualizing pixels inside a patch as its measurements, the problem is naturally a spatio-time sequence prediction. This spatio-time sequence prediction problem is different from that of one-step time series prediction because this prediction target contains both spatial and temporal structures.

CNNs use very little pre-processing: they learn the filters, hand-engineered in conventional algorithms.

The objective is to maximize the size K of $\mathbf{E}_K$ set.

The study presents the experiments carried out to compare the performance of this proposed deep learning model, tested with the dataset, introduced in the following subsection, which was preprocessed.

The Root Mean Squared Error (RMSE), the mean absolute error (MAE), the mean square error (MSE) and R-squared (R^2) are the measures used to assess model performance across all experiments. Since the F score is derived from recall and precision, the study also shows these two measures for reference. The results are presented, discussed, and analyzed in the following sections.

An excellent monitoring economic environment is needed to collect messages from an upcoming economic crisis event. To verify the performance of the proposed model, the study applied three evaluation indices, including the mean squared error (RMSE), the mean absolute error (MAE), and goodness of fit (R-Square) as the loss function for model training.

The study divided the data sets into a training set and a verification set. The learning set is applied to train different deep-learning models while updating the weights and biases of the neural cell. Then the verification set checks the skill of these models.

LSTM is an important part of the CNN-LSTM framework and provides vector characteristics based on historical information.

The research study checked the effectiveness of the proposed CNN-LSTM-based deep learning model against the benchmarks: the RNN and LSTM prediction methods are the widely used deep learning models.